# Marketing Analytics

## Chapter 6 · Mechanisms & Heterogeneity

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# Before we start

**Paper quiz on Chapter 5, Part II** *(§5.7–5.10, 8 min)*

- Individual, closed book
- Counts toward participation

**Assignment 3** is running

- Deadline: Group A Sun, Nov 22 · Group B Wed, Nov 25
- Debrief at the start of next session

# Today's session

6.1 The average effect hides the interesting part
6.2 Interactions and **marginal effects**
6.3 Continuous × categorical
6.4 Continuous × continuous *(short)*
6.5 Categorical × categorical: a **2 × 3 experiment**
6.6 Pitfalls: subgroup fishing & power
6.7 **Mediation**: why does it work?
6.8 Good and bad controls *(short)*
6.9 Your group project

> **Driving question:** *Why — and for whom — does it work?*
>
> One new ingredient: the **product of two variables**.

# 6.1 The average effect hides the interesting part

# Smart Mix, second wave

- **1,200 users**, randomized to Smart Mix or control
- This time with **covariates**: plan (free/Premium), prior listening, tenure, device
- And one more variable: **new artists discovered** during the four weeks
- Average effect: **+3.7 hours** (95% CI 3.1–4.3)

The product team asks two more questions:

1. **For whom** does it work — free or Premium users? light or heavy listeners?
2. **Why** does it work — does it make users discover new music?

**Moderation** (for whom): the effect *depends on* a third variable. **Mediation** (why): the effect *runs through* a third variable. Easy to confuse — a moderator changes *how large*, a mediator is *part of the path*.

# 6.2 Interactions and marginal effects

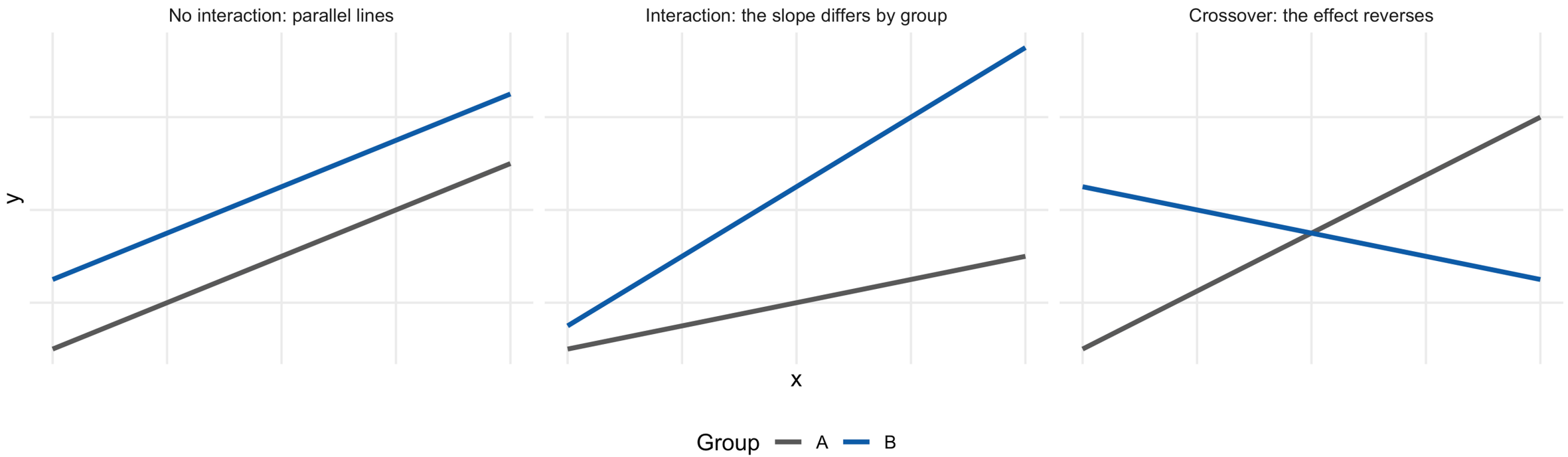
# What an interaction term does

$$y = \beta_0 + \beta_1 x + \beta_2 z + \beta_3 (x \times z) + \varepsilon \qquad \Rightarrow \qquad \frac{\partial y}{\partial x} = \beta_1 + \beta_3 z$$

- The **marginal effect** of $x$ depends on $z$
- $\beta_1$ = effect of $x$ **when** $z = 0$ (a *conditional* effect, not a "main effect")
- $\beta_3$ = how much the effect of $x$ changes per unit of $z$
- Symmetric: effect of $z$ = $\beta_2 + \beta_3 x$

**Two rules:** (1) if zero is not a meaningful value of $z$, $\beta_1$ is not meaningful either → center (§6.4). (2) **Always keep both main terms**: in R, `x * z` = `x + z + x:z`.

# Parallel, diverging, crossing



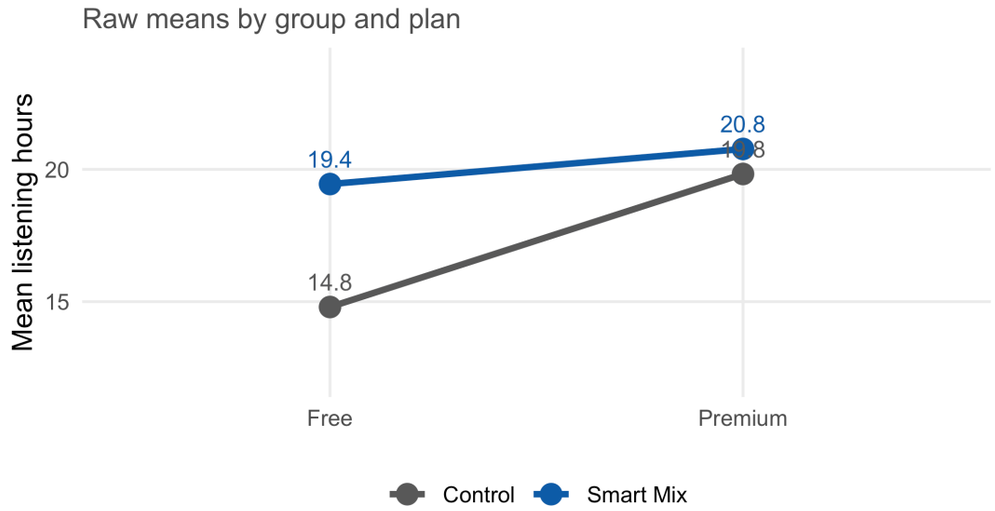

With a group indicator as $z$: **no interaction** = parallel lines (same slope, different level) · **interaction** = different slopes · **crossover** = the effect of $x$ reverses. $\beta_3$ is simply the **difference between the slopes**.

# 6.3 Continuous × categorical

# Face 1: a treatment effect that differs between groups

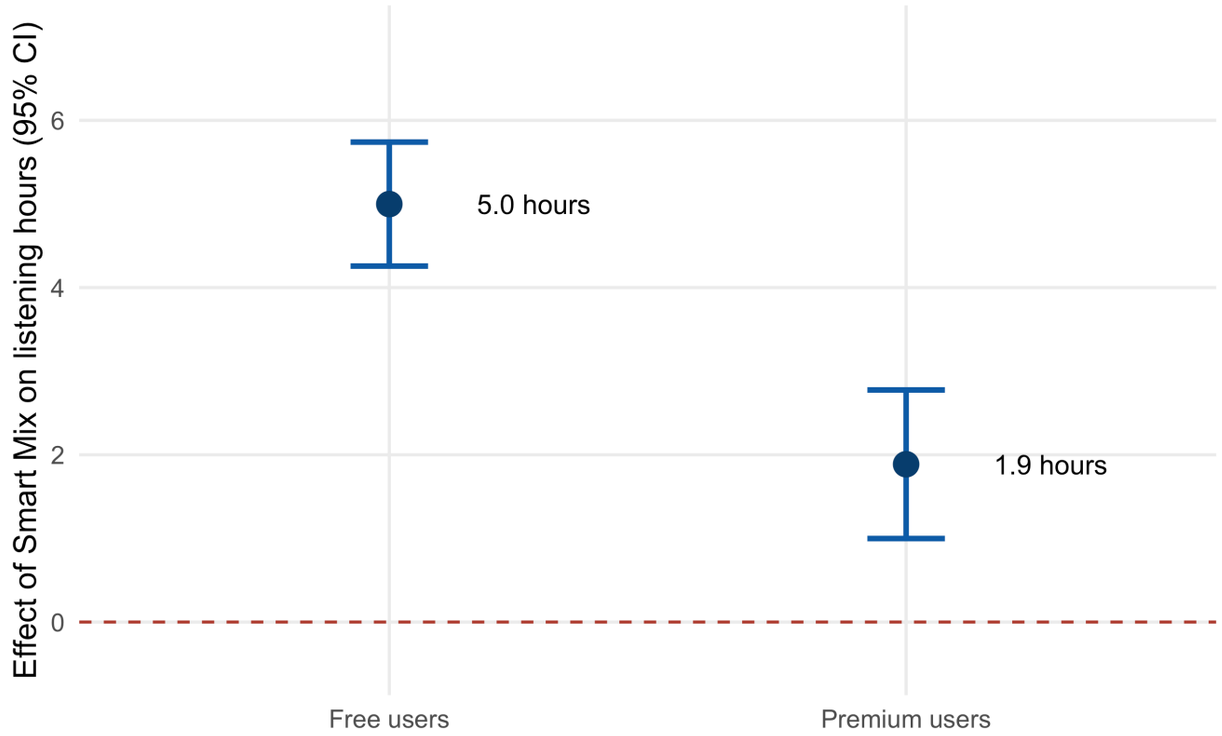
```r
model_plan <- lm(hours ~ smart_mix * premium + prior_hours + tenure, data = wave2)
coef_table(model_plan)
```

| | Estimate | Std. Error | t value | p-value |
|---|---|---|---|---|
| (Intercept) | 1.560 | 0.469 | 3.33 | < 0.001 |
| smart_mix | 4.999 | 0.378 | 13.21 | < 0.001 |
| premium | 2.685 | 0.431 | 6.22 | < 0.001 |
| prior_hours | 0.752 | 0.017 | 43.97 | < 0.001 |
| tenure | 0.028 | 0.009 | 3.26 | 0.001 |
| smart_mix:premium | −3.111 | 0.590 | −5.27 | < 0.001 |




`smart_mix` = effect **for free users** (premium = 0): **5.0 h** · `smart_mix:premium` = how much the effect **differs** for Premium: **−3.1 h** (the heterogeneity test) · effect for Premium = 5.0 − 3.1 = **1.9 h** · `premium` = plan difference **in the control group**.

# Both effects with their uncertainty



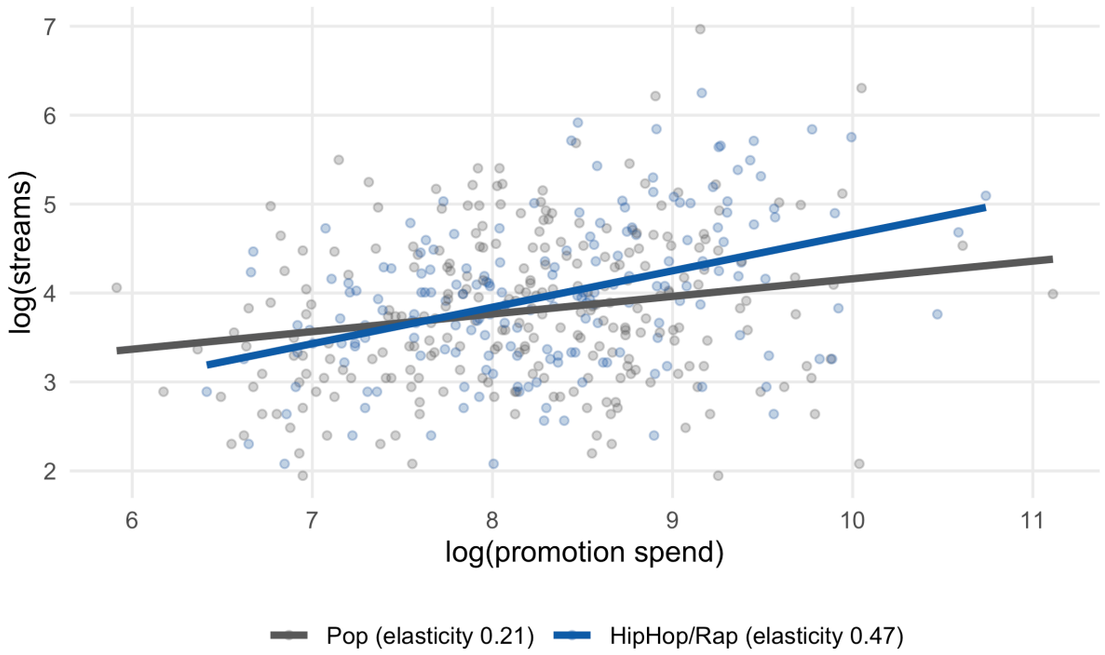

- The CI of the Premium effect is **not** in the output — it combines two coefficients and their covariance (`emmeans` does it for you)
- Actionable: promote Smart Mix to **free users first**

> The interaction coefficient tests whether the effects **differ**; the two marginal effects tell you **what they are**. Report both.

# Face 2: a slope that differs between groups

```r
model_genre <- lm(log(streams) ~ log(spend) * genre + log(followers) + plac
                  data = promo)
coef_table(model_genre)
```

| | Estimate | Std. Error | t value | p-value |
|---|---|---|---|---|
| (Intercept) | −3.648 | 0.321 | −11.36 | < 0.001 |
| log(spend) | 0.206 | 0.033 | 6.17 | < 0.001 |
| genreHipHop/Rap | −2.058 | 0.430 | −4.79 | < 0.001 |
| genreRock | 0.356 | 0.654 | 0.54 | 0.587 |
| genreElectro/Dance | 0.021 | 0.616 | 0.03 | 0.973 |
| log(followers) | 0.507 | 0.015 | 33.46 | < 0.001 |
| placements | 0.122 | 0.011 | 11.42 | < 0.001 |
| log(spend):genreHipHop/Rap | 0.268 | 0.052 | 5.18 | < 0.001 |
| log(spend):genreRock | −0.053 | 0.079 | −0.67 | 0.504 |
| log(spend):genreElectro/Dance | 0.005 | 0.076 | 0.07 | 0.946 |




Same model, different question: `log(spend)` = elasticity for **Pop** (reference); `log(spend):genreHipHop/Rap` = **+0.27** → HipHop/Rap elasticity ≈ **0.47**. Rock and Electro: no evidence of a different slope. Genre "main effects" = level differences at **1 euro** of spend.

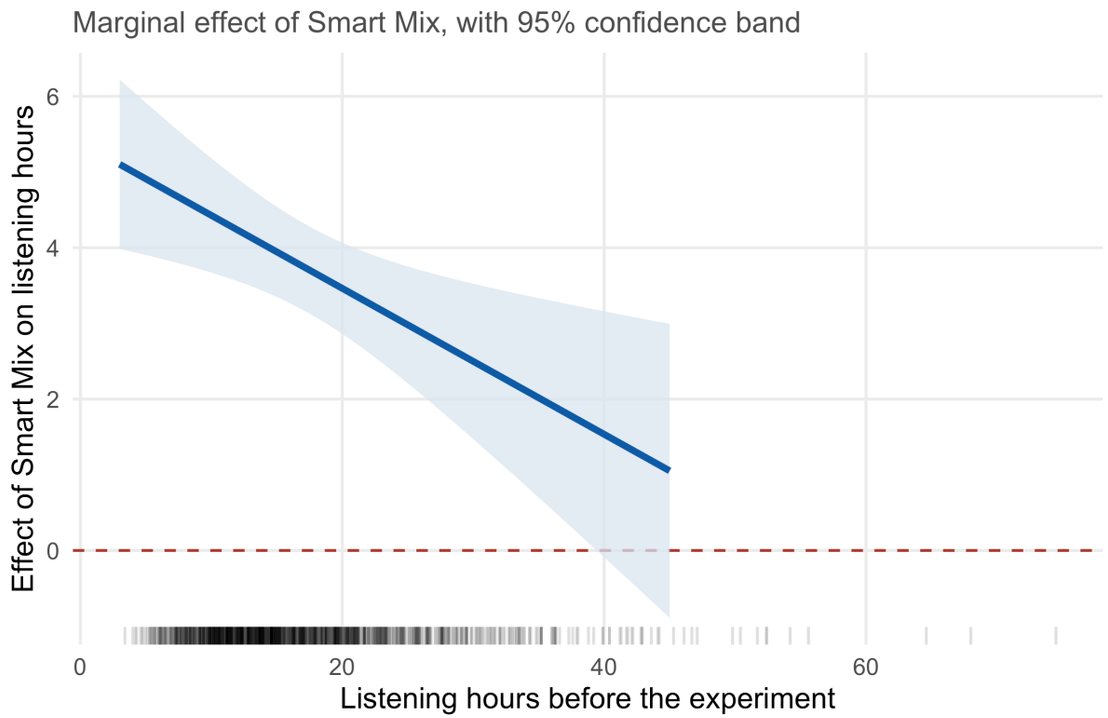
# Moderation by a continuous variable

```
model_prior <- lm(hours ~ smart_mix * prior_hours + premium + tenure, data
coef_table(model_prior)
```

| | Estimate | Std. Error | t value | p-value |
|---|---|---|---|---|
| (Intercept) | 1.446 | 0.532 | 2.72 | 0.007 |
| smart_mix | 5.391 | 0.660 | 8.16 | < 0.001 |
| prior_hours | 0.797 | 0.023 | 34.43 | < 0.001 |
| premium | 1.041 | 0.301 | 3.46 | < 0.001 |
| tenure | 0.027 | 0.009 | 3.15 | 0.002 |
| smart_mix:prior_hours | −0.096 | 0.034 | −2.82 | 0.005 |

Marginal effect of Smart Mix = 5.39 − 0.096 × prior hours:

- 10 hours before → **4.4 h** · 30 hours → **2.5 h**
- `smart_mix` alone = effect at **zero** prior hours — a user who hardly exists




With a continuous moderator, report the effect at **meaningful values** (10th / 50th / 90th percentile) or as a **curve with its confidence band** — never the coefficient at zero.

# Your turn (5 min)

Fit the interaction of Smart Mix with **device**:

```
model_device <- lm(hours ~ smart_mix * device + prior_hours + premium + tenure, data = wave2)
coef_table(model_device)
```

For which device is the effect of Smart Mix largest? Compute the effect for each device from the coefficients. Are the differences significant?

# 6.4 Continuous × continuous

# Spend × followers: the problem with “main effects”

**Uncentered:**

```
m_cc <- lm(log(streams) ~ log(spend) * log(followers) +
             placements + genre, data = promo)
coef_table(m_cc)[2:3, ]
```

```
               Estimate Std. Error t value p-value
log(spend)       -0.896      0.183   -4.90 < 0.001
log(followers)   -0.422      0.143   -2.96   0.003
```

A **negative** spend elasticity? It is the elasticity for an artist with **one follower** (log = 0).

**Centered at the means:**

```
promo_c <- promo |> mutate(ls = log(spend) - mean(log(spend)),
                           lf = log(followers) - mean(log(followers)))
m_ccc <- lm(log(streams) ~ ls * lf + placements + genre, data = promo_c)
coef_table(m_ccc)[c(2, 3, 8), ]
```

```
      Estimate Std. Error t value p-value
ls       0.299      0.022   13.31 < 0.001
lf       0.508      0.015   33.87 < 0.001
ls:lf    0.113      0.017    6.53 < 0.001
```

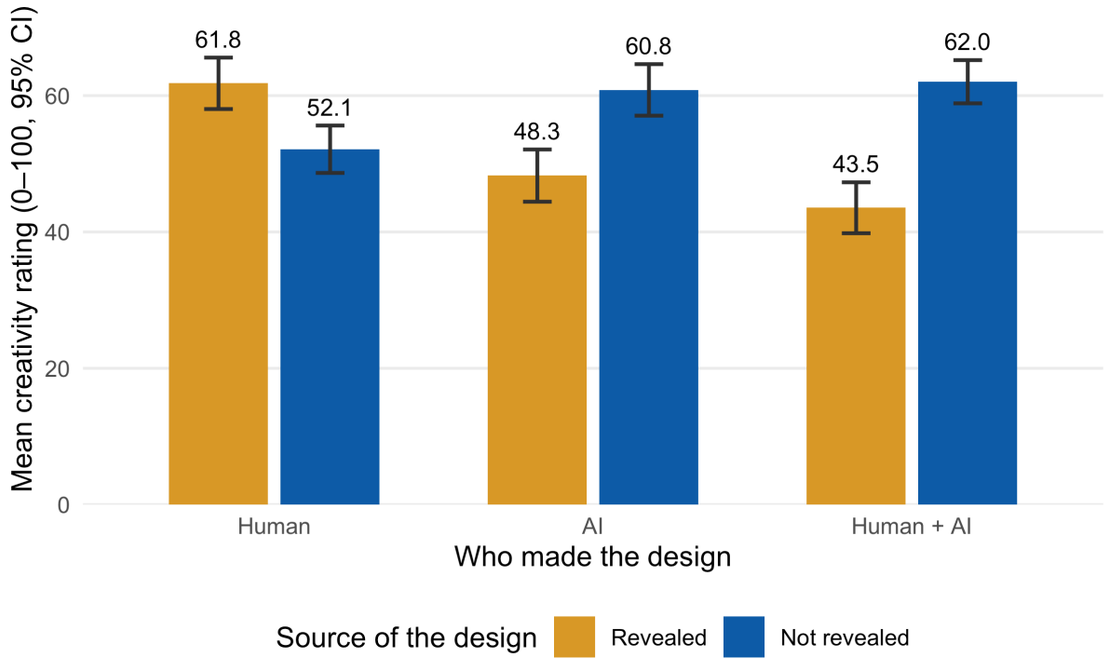
Now `ls` = elasticity for an **average** artist; the interaction is unchanged.

> Centering changes the main effects (now “at the mean of the other variable”), **not** the interaction or the fit. Communicate with a **plot of the marginal effect** — book §6.4 — not with the table.

# 6.5 Categorical × categorical: a 2 × 3 experiment

# Are AI-made ads judged as creative as human-made ads?

- **360 respondents** rate the creativity of an ad (0–100)
- Factor 1 — **source** of the design: Human · AI · Human + AI
- Factor 2 — **disclosure**: source revealed · not revealed
- 2 × 3 = **6 cells**, 60 respondents each, randomized




Revealed: human designs win. Not revealed: AI and hybrid designs are rated **as creative** — slightly higher. **Crossing lines = the effect of the source depends on disclosure.**

# The regression: six coefficients, six cell means

```
model_ads <- lm(creativity ~ source * disclosure, data = ads)
coef_table(model_ads)
```

| | Estimate | Std. Error | t value | p-value |
|---|---|---|---|---|
| (Intercept) | 61.800 | 1.854 | 33.33 | < 0.001 |
| sourceAI | −13.533 | 2.622 | −5.16 | < 0.001 |
| sourceHuman + AI | −18.250 | 2.622 | −6.96 | < 0.001 |
| disclosureNot revealed | −9.667 | 2.622 | −3.69 | < 0.001 |
| sourceAI:disclosureNot revealed | 22.233 | 3.708 | 6.00 | < 0.001 |
| sourceHuman + AI:disclosureNot revealed | 28.150 | 3.708 | 7.59 | < 0.001 |

| Coefficient | Meaning (reference: Human, Revealed) |
|---|---|
| `(Intercept)` 61.8 | Human designs, source revealed |
| `sourceAI` −13.5 · `sourceHuman + AI` −18.3 | AI / hybrid designs, *when revealed*, vs. human |
| `disclosureNot revealed` −9.7 | *Human* designs: not revealed vs. revealed |
| `sourceAI:…` +22.2 · `sourceHuman + AI:…` +28.2 | The **extra** gain of AI / hybrid designs when not revealed, beyond the two main effects |

# Step 1: is there an interaction at all?

```
1 anova(model_ads)
```

```
Analysis of Variance Table

Response: creativity
                  Df Sum Sq Mean Sq F value  Pr(>F)
source             2   1055     527    2.56   0.079 .
disclosure         1   4572    4572   22.17 3.6e-06 ***
source:disclosure  2  13218    6609   32.04 1.6e-13 ***
Residuals        354  73021     206
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

> The ANOVA table tests each factor and the **interaction as a whole** (two coefficients at once). Interaction highly significant → the effect of the source depends on disclosure → **do not interpret the main effects on their own**. (Unbalanced cells: `car::Anova(model_ads, type = 2)`.)

# Step 2: which cells differ? Estimated marginal means

```r
library(emmeans)
emm <- emmeans(model_ads, ~ source | disclosure)
pairs(emm)
```

```
disclosure = Revealed:
 contrast             estimate   SE  df t.ratio p.value
 Human - AI              13.53 2.62 354   5.161  <.0001
 Human - (Human + AI)    18.25 2.62 354   6.960  <.0001
 AI - (Human + AI)        4.72 2.62 354   1.799  0.1715

disclosure = Not revealed:
 contrast             estimate   SE  df t.ratio p.value
 Human - AI              -8.70 2.62 354  -3.318  0.0029
 Human - (Human + AI)    -9.90 2.62 354  -3.775  0.0005
 AI - (Human + AI)       -1.20 2.62 354  -0.458  0.8910

P value adjustment: tukey method for comparing a family of 3 estimates
```

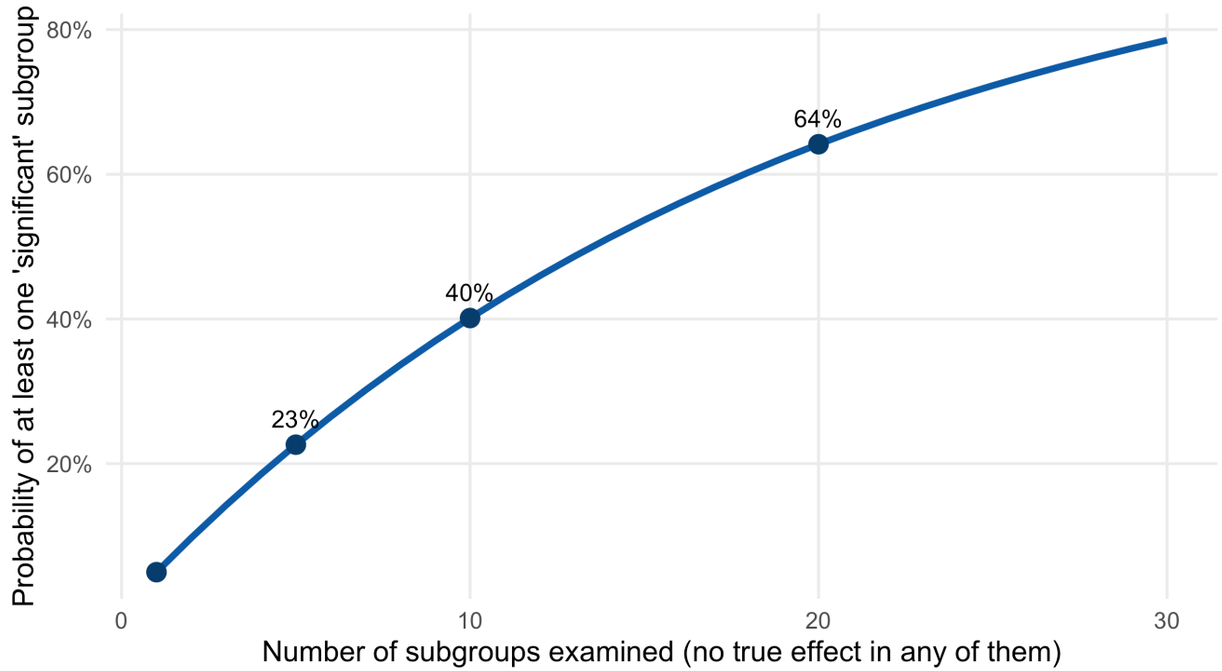
> From “there is an interaction” to “**where exactly** are the differences”: revealed → human beats AI (14 points) and hybrid; not revealed → AI and hybrid beat human by 9–10 points. *p*-values **Tukey-adjusted** within each disclosure condition — the two-step logic of Chapter 4, with one extra dimension.

# Your turn (5 min)

1. Rebuild the cell mean for **hybrid** designs with the source **not revealed** from the coefficients of `model_ads`. Check it against the table of cell means.
2. Which **single number** in the regression output answers: “does disclosure change the rating of hybrid designs *more* than it changes the rating of human designs?”

# 6.6 The pitfalls of looking for heterogeneity

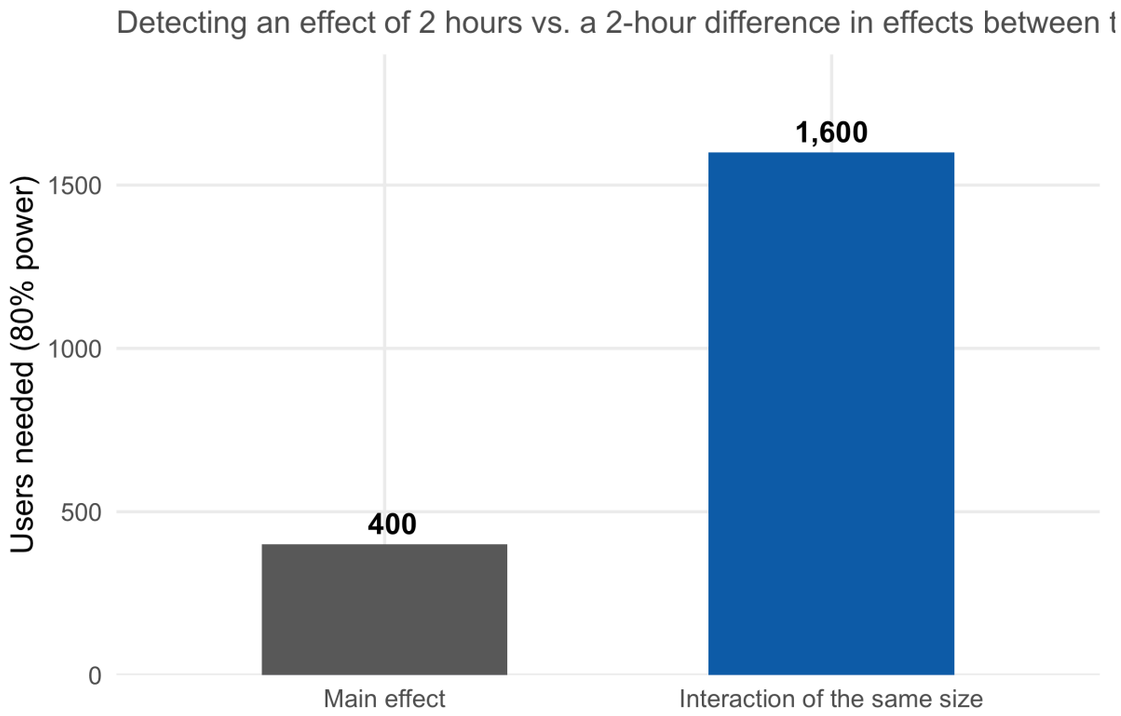
# Subgroup fishing




- Plan × device × tenure × age × country … = dozens of subgroups
- Testing each = **multiple testing** (Chapter 4)
- 20 subgroups, no heterogeneity: **64%** chance of a "significant" one
- Live demo: subgroup-fishing widget, book §6.6

> "Smart Mix works particularly well for Android users aged 25–34" after twenty subgroups = **noise with a story**.

# Interactions need large samples




- SE of a *difference between two effects* = **2 ×** SE of the overall effect
- Same power → **4 × the sample** for an interaction as large as a main effect
- And interactions are usually **smaller**
- Underpowered: miss real heterogeneity, **exaggerate** what you find

**Pre-register** the moderators you expect and test those. Everything else is exploratory — a hypothesis for the *next* experiment. Not significant → “no evidence that the effect differs”, not “the effect is the same”.

# 6.7 Mediation: why does it work?

# Total = direct + indirect

# Three regressions

```r
model_m <- lm(new_artists ~ smart_mix + prior_hours + premium, data = wave2)          # path a
model_y <- lm(hours ~ smart_mix + new_artists + prior_hours + premium + tenure, data = wave2)   # paths b and c'
a <- coef(model_m)["smart_mix"]; b <- coef(model_y)["new_artists"]; c_direct <- coef(model_y)["smart_mix"]
c(a = unname(a), b = unname(b), direct = unname(c_direct), indirect = unname(a * b), total = unname(c_direct + a * b))
```

```
     a        b   direct indirect    total
3.6185   0.3909   2.3063   1.4146   3.7209
```

- **a** = 3.6 more new artists with Smart Mix
- **b** = 0.39 hours per new artist, holding Smart Mix constant
- **c′** = 2.31 hours direct — down from the total 3.72

> Indirect effect a × b = **1.42 hours = 38%** of the effect runs through discovery. Discovery is **an** important mechanism — not the only one.

# Uncertainty of the indirect effect: bootstrap

```r
set.seed(1)
boot_indirect <- replicate(2000, {
  d_boot <- wave2[sample(nrow(wave2), replace = TRUE), ]
  coef(lm(new_artists ~ smart_mix + prior_hours + premium, data = d_boot))["smart_mix"] *
    coef(lm(hours ~ smart_mix + new_artists + prior_hours + premium + tenure, data = d_boot))["new_artists"]
})
quantile(boot_indirect, c(0.025, 0.975))
```

```
  2.5%  97.5% 
0.9861 1.8632 
```

> A product of two estimates is **not normally distributed** → no textbook standard error → **bootstrap** (Chapter 3): resample users, re-estimate, look at the distribution. Report **total, direct, indirect** with uncertainty, plus the **share mediated**. (`mediation` package: `mediate()` does this incl. logistic outcomes.)

# The catch: nobody randomized the mediator

Live demo: book §6.7 — mediation widget, switch on the **unobserved confounder of M and Y**

https://wu-rds.github.io/MA2026/06-mechanisms.html#sec-mech-mediation

*Screenshot placeholder: the mediation widget*
`img/ch06/widget_mediation.png`

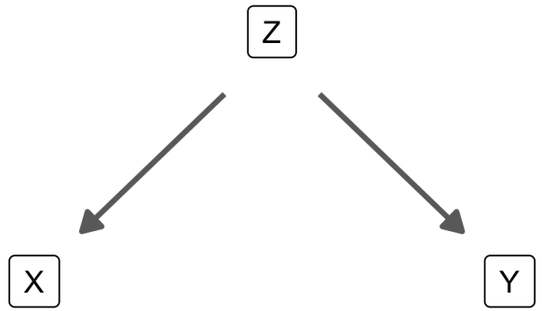
> Randomization identifies the **total** effect. The **split** needs one more assumption: **no unobserved confounder of mediator and outcome** (curious users discover more *and* listen more). Untestable — and the same problem as Chapter 5, one step down the chain.

What helps: measure likely confounders of M; **manipulate the mediator** (Smart Mix with vs. without discovery).

# 6.8 Good and bad controls

# Fork, pipe, collider

**Fork (confounder)**



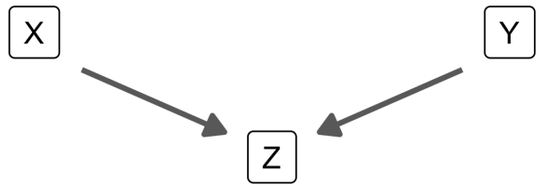

adjust for Z

**Pipe (mediator)**




do NOT adjust for Z
(unless you want the direct effect)

**Collider (common effect)**




do NOT adjust for Z

**Fork** = confounder → adjust (Ch. 5). **Pipe** = mediator → adjusting gives the *direct* effect, not the total. **Collider** = common effect → adjusting (or selecting on it) **creates** a spurious association. Draw the diagram **before** the analysis.

# 6.9 Your group project

# Moderation and mediation with 120 respondents

## Add a moderator?

- Only if central to your topic and you can state the **expected direction** in advance
- Measure it **before** the manipulation
- `outcome ~ condition * moderator`
- 60 per condition → only **large** interactions detectable
- A 2 × 2 spreads you over cells of 30

## Measure a mediator?

- One or two plausible mechanisms, measured **after** the manipulation (perceived quality, trust, attention)
- Cheap — and tells you **where the chain broke** if the manipulation did not work
- Report total / direct / indirect **with the caveats**

> State what was **planned** and what was **exploratory**. “No evidence that the effect differs” ≠ “the effect is the same”.

# Learning check

1. `smart_mix` = 5.0, `smart_mix:premium` = −3.1. Effect for free users? For Premium? What does the interaction's *p*-value test?
2. 2 × 2: intercept 40, `framingYearly` −5, `trialYes` +8, `framingYearly:trialYes` +6. The four cell means? Where does the trial help more?
3. A loyalty program's total effect on spending, with "store visits" as a control — fork, pipe, or collider? Consequence?
4. 25 segments tested, "works particularly well for returning customers on tablets" (*p* = .02). Problem — and what next?

# Next session

**Module 8 · Survey design** — *from research question to questionnaire*

- Measuring constructs: items, scales, reliability
- Embedding your experiment in a Qualtrics survey
- Questionnaire structure, pretesting, data collection

> Bring your group's **research question and planned manipulation** — we will work on your questionnaires in class.